Check for updates

Proceedings of the ASME 2024 Heat Transfer Summer Conference
HT2024
July 15-17, 2024, Anaheim, California

HT2024-131027

Downloaded from http://asmedigitalcollection.asme.org/HT/proceedings-pdf/HT2024/87905/V001T03A003/3754451/v001t03a003-ht2024-131027.pdf by Tsinghua University user on 11 March 2026

PREDICTION OF CRITICAL HEAT FLUX FOR LIQUID HELIUM USING MACHINE LEARNING MODELS ASSISTED BY PHYSICS-BASED CORRELATION

Jiayuan Li[1,*] **Chirag R.Kharangate[1]**

[1]Case Western Reserve University, Cleveland, OH

ABSTRACT

Liquid helium plays an important role in fields such as superconductivity, quantum computing, medical cryopreservation, and space exploration. Prediction of pool boiling heat transfer, especially the critical heat flux, is crucial for all these applications. To date, no effective tool is available to give accurate predictions of critical heat flux specific for liquid helium. In this study, we use both traditional machine learning models and physics-assisted machine learning models to predict critical heat flux of liquid helium with 585 data sourced from 22 studies. While the traditional machine learning models possess no information on critical heat flux, the physics-assisted machine learning models obtain physical information of critical heat flux from Zuber [2]'s correlation by predicting the multiplier of the hydrodynamic instability term. Recursive feature elimination and stepwise selection are used to remove the redundant features while multivariate adaptive regression splines, support vector machine, random forest, and extreme gradient boosting are used for modeling. Support vector machine, random forest, and extreme gradient boosting result in better performance than traditional correlations. Physics-assisted extreme gradient boosting model achieves the best result with 14% mean absolute percentage error. This work is the first attempt in reported literatures to predict critical heat flux for liquid helium using machine learning models.

Keywords: pool boiling; machine learning; critical heat flux; liquid helium

NOMENCLATURE

Roman letters

CHF	critical heat flux, [W/m^2]
C_p	heat capacity at constant pressure, [J/(kgK)]
D	diameter of disk plate, [m]
h_{fg}	latent heat of evaporation, [J/kg]
HTC	heat transfer coefficient, [W/(m^{2}K)]
K	multiplier of Zuber's model
k	heat conductivity, [W/(mK)]
MARS	multivariate adaptive regression
MAPE	mean absolute percentage error
P	pressure, [Pa]
P_{cr}	critical pressure, [Pa]
P_r	reduced pressure, $P_r = P/P_{cr}$
Pr	Prandtl number, $Pr = \mu C_p/k$
q''	heat flux, [W/m^2]
RF	Random forest
RMSE	Root mean square error
SVM	support vector machine
T	temperature, [K]
XGBoost	extreme gradient boosting

Greek letters

θ	orientation of plate, [°]
μ	dynamic viscosity, [Pa·s]
λ	wavelength in vapor-liquid interface, [m]
λ_t	Taylor's instability wavelength, [m]
$\lambda_{t,d}$	most dangerous Taylor's instability, [m]
λ_h	Helmholtz unstable wavelength, [m]
ρ	density, [kg/m^3]
σ	surface tension, [N/m]

Subscripts

atm	atmospheric
exp	experimental
f	liquid
g	vapor
pred	predicted
sat	saturated
sub	subcooled

1. INTRODUCTION

As one of the most effective heat transfer processes, pool boiling transfers heat through phase change between liquid and vapor on a static pool so that latent heat plays a role in dramatically increasing the heat transfer by sensible heat. Liquid

Copyright © 2024 by ASME

helium (LHe) is one of the cryogenic fluids with extremely low boiling point ($T_{sat,atm}$ = 4.22 K) and behaves as a superfluid (μ_f = 0) at temperatures below 2.17 K. Applications of LHe include superconducting magnets cooling, quantum computing, cryopreservation for medical samples, and infrared telescope for space exploration etc. All these applications involve pool boiling heat transfer management, and therefore accurate prediction of pool boiling heat transfer process of LHe becomes highly crucial. A pool boiling curve is an effective tool to estimate the heat transfer process that can be divided into four boiling regimes connected by three transitional points as shown on Fig. 1. Among them, critical heat flux (CHF) plays a key role for its correspondence with maximum heat flux as well as point of boiling crisis where boiling surface could melt due to overheating. Therefore, accurate prediction of CHF becomes a priority on LHe.

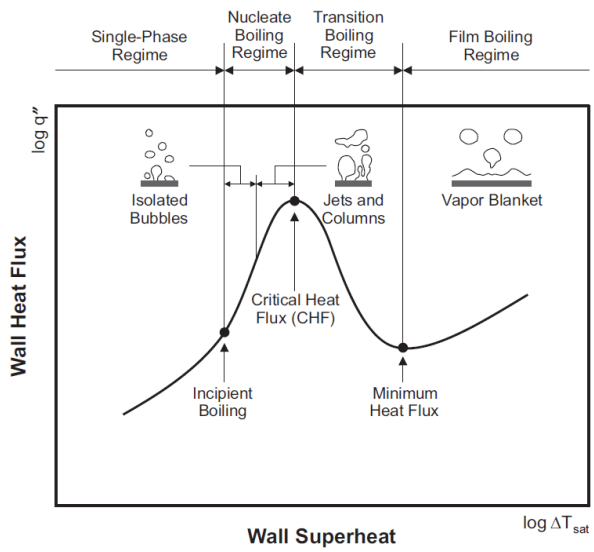


FIGURE 1: POOL BOILING CURVE [1].

Semi-empirical correlations are the traditional tools used to estimate the pool boiling CHF. The hydrodynamic instability model of Zuber [2] is one of the most popular models for its reasonable and stable predictions on CHF across many types of fluids, and ability to incorporate experimental data under different operating conditions to improve the predictive accuracy. Some of the efforts include Wang et al. [3], Bewilogua et al. [4] for pressure, Chang and You [5] for orientation, and Ivey and Morris [6] for subcooling. However, the predictive power of many of the correlation models including Zuber [2] are limited by their poor regression capability over relevant parameters, as all the relevant parameters are nonlinearly related to CHF and among each other.

Recently, machine learning (ML) models have been increasingly used as a more accurate tool on two-phase heat transfer management compared with traditional correlations. Qiu et al. [7] developed an artificial neural network model with seven hidden layers to predict heat transfer coefficient (HTC) of saturated flow boiling in mini/micro channels. The results show a 14.3% mean absolute percentage error compared with 27.4% of the universal correlation of Kim and Mudawar [8] for flow boiling HTC. Zhou et al. [9] applied artificial neural network, random forest, adaptive boosting, and extreme gradient boosting models to predict flow condensation HTC in mini/micro channels and the best results are achieved by artificial neural network with 6.8% mean absolute error, which is far better than 26.6% of the universal model of Kim and Mudawar [10]. Furthermore, Bahman and Ebrahim [11] developed an artificial neural network model composed of 12 neurons inside 2 hidden layers to predict the minimum film boiling temperature on a distilled water pool, and the trained model shows only 5% error. Ravichandran and Bucci [12] successfully applied the one-layer feed-forward artificial neural network to quantify the bubble growth time, bubble period, and nucleation site density with high accuracy. Kim et al. [13] combined the traditional correlation models with multi-layer perception and random forest model and concluded that the correlation assisted ML models are capable of better predictions than any stand-alone models.

The objective of this study is to find the most accurate ML model for predicting pool boiling CHF of LHe. To achieve the goal, a physics-assisted machine learning (PAML) framework is introduced in which we use physics-based correlation to promote and assist the prediction of traditional data-driven machine learning (DDML) models. Multivariate adaptive regression, support vector machine, random forest, extreme gradient boosting are applied on both frameworks for modeling. As an important preparation before modeling, recursive feature elimination and stepwise feature selection are used to remove redundant features. Finally, the predictions of DDML, PAML, and traditional correlations are compared, and the best performed model is selected as the most accurate tool to predict pool boiling CHF of LHe.

2. PREPARATION AND METHODS FOR MODELING

2.1 Exploratory data analysis

CHF of LHe in current study are consolidated from worldwide published literatures [4,16−36] that includes wide range of operational conditions and thermophysical properties. Table 1 includes important information of individual studies throughout the consolidated dataset. We note that all the input features are thermophysical properties of LHe that exist as important parameters of the traditional semi-empirical correlations for pool boiling CHF [14], which are: ρ_g, ρ_f, h_{fg}, σ, P, Pr_f, Pr_g C_{pf}, T_{sat}, k_f, μ_f, μ_g, ΔT_{sub}, θ.

In this study, R studio is used as the main tool for statistical data exploration and modeling, which serves as an integrated development environment (IDE) for R programming language that is designed for statistic computing, graphic visualization, and machine learning. The thermophysical properties of LHe are

Downloaded from http://asmedigitalcollection.asme.org/HT/proceedings-pdf/HT2024/87905/V001T03A003/7375451/v001t03a003-ht2024-131027.pdf by Tsinghua University user on 11 March 2026

Copyright © 2024 by ASME

collected from NIST's REFPROP 9.0 [15] software based on the specific operational conditions of each observation.

TABLE 1: CONSOLIDATED DATASET OF LHE

Authors	Pressures [MPa]	Orientation [°]	Surface Material (Size: Width × Length [mm^2] or Diameter [mm])	Datapoints
Lyon [16]	0.005 – 0.217	0°,45°,65°,90°,135°, 160°,180°	Platinum (9.91)	48
Cummings and Smith [17]	0.101	0°	Copper (15.24)	17
Jergel and Stevenson [18]	0.101	0°,90°,180°	Copper (15)	3
Jergel and Stevenson [19]	0.101	0°,90°,180°	Aluminum 59 (15)	3
Jergel and Stevenson [20]	0.101	0°,90°,180°	Aluminum 69 (15)	3
Tsukamoto and Kobayashi [21]	0.101	0°	Stainless steel (37)	2
Bewilogua et al. [4]	0.006 – 0.218	0°,90°,135°,165°	Copper (24.97)	100
Vishnev et al. [22]	0.101	0°,30°,45°,60°,90°,120°, 150°	Stainless steel (96 × 10.4)	8
Deev et al. [23]	0.1 – 0.226	0°,90°	Copper (30 × 30)	40
Grigoriev et al. [24]	0.101	0°	Copper (16)	7
Ibrahim et al. [25]	0.098 – 0.13	0°	Copper (25.4)	12
Tsuruga and Endoh [26]	0.009 – 0.223	0°	Copper (20)	35
Verkin et al. [27]	0.1 – 0.2	0°	Copper (20)	32
Ogata and Nakayama [28]	0.101	0°	Copper (15 × 15)	2
Caspi [29]	0.121	90°	Copper (12.716)	4
Nishio and Chandratilleke [30]	0.101	0°,45°,90°,135°, 175°	Copper (20)	8
Ogata and Mori [31]	0.101	0°,90°	Copper (15 × 15)	2
Chandratilleke et al. [32]	0.101	0°,90°,180°	Copper (20)	17
Iwamoto et al. [33]	0.101	0°,15°,30°,45°,60°,75°,90°,105°,120°,135°,150°,165°,172°	Copper (18 × 76)	64
Iwamoto et al. [34]	0.101	0°,15°,30°,45°,60°,75°,90°,105°,120°,135°,150°,165°,180°	Copper (18 × 10, 18 × 18, 18 × 40, 18 × 76)	51
Hata et al. [35]	0.101 – 0.13	0°,90°,135°,180°	Manganin (10 × 40)	121
Iwamoto et al. [36]	0.101	0°	Copper (20)	6

Fig. 2 illustrates the distribution of datapoints of the consolidated dataset over reduced pressure (P_r) (Fig. 2(a)) and orientation (θ) (Fig. 2(b)). Fig. 2(a) shows that datapoints are more skewed towards atmospheric pressure than any other pressures. Fig. 2(b) shows more datapoints are skewed towards upward-facing ($\theta = 0°$) compared to downward-facing ($\theta = 180°$) position for flat plate. As an important preparation before modeling, the dataset is subdivided into subsets of 70% for training and 30% for testing. We should note that the partition has been carefully done so that the training and testing datasets share the same structure with the complete dataset over all the important features (pressure, orientation etc.).

For testing, we use both the mean absolute percentage error (MAPE) and the root mean square error (RMSE). MAPE is defined as error in percentage while RMSE is with the same unit as the targeted feature. Equation (1) defines MAPE and Equation (2) is RMSE.

$$MAPE = \frac{1}{N}\sum \frac{\left|q''_{CHF,pred} - q''_{CHF,exp}\right|}{q''_{CHF,exp}} \times 100\% \quad (1)$$

$$RMSE = \sqrt{\frac{\sum\left(q''_{CHF,pred} - q''_{CHF,exp}\right)^2}{N}} \quad (2)$$

Copyright © 2024 by ASME

Downloaded from http://asmedigitalcollection.asme.org/HT/proceedings-pdf/HT2024/87905/V001T03A003/3754451/v001t03a003-ht2024-131027.pdf by Tsinghua University user on 11 March 2026

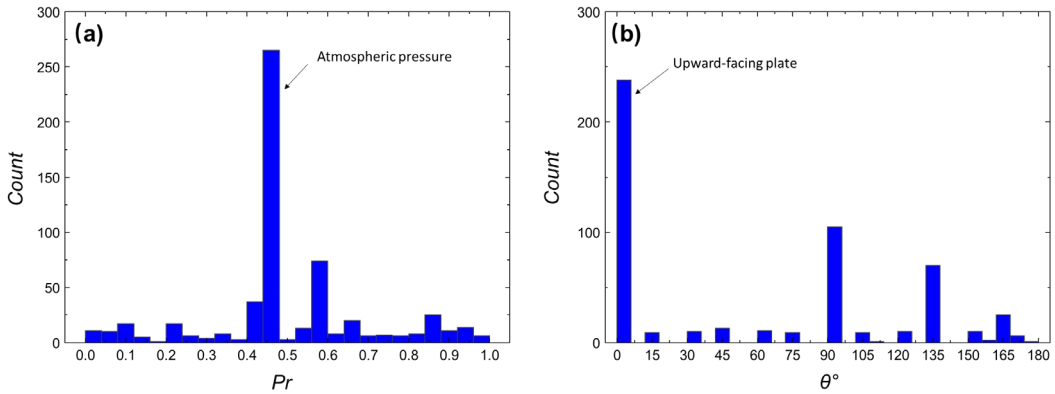


FIGURE 2: STRUCTURE OF THE DATASET.

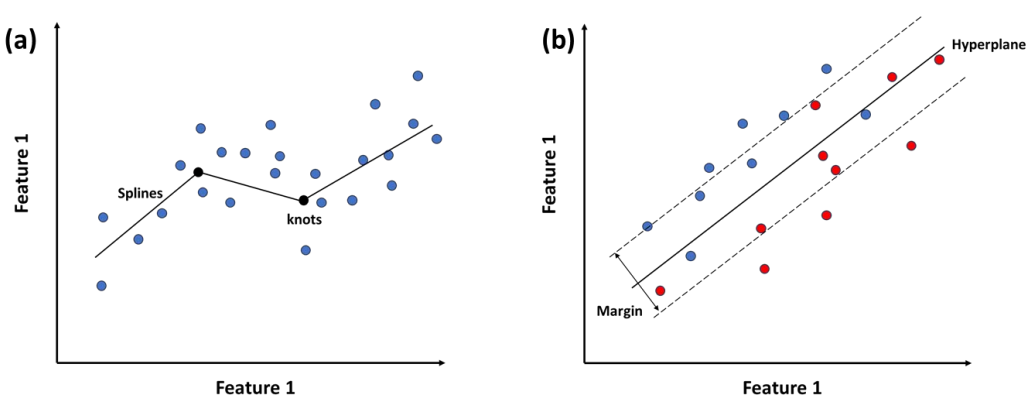
2.2 Feature selection methods

In terms of modeling, redundant features refer to the input features that provide only trivial contribution to the modeling while negatively impact modeling by overfitting. In this study, we use recursive feature elimination (RFE) and stepwise selection (SWS) methods to remove the redundant features while keeping as much information as possible before developing any machine learning models.

The basic idea of RFE is to use the selected ML model on a given train dataset to find the contribution of each feature to modeling. This process happens recursively until all the irrelevant features are identified or removed, and the selected model reaches maximum accuracy. In this study, we use RF model to train on the dataset, and all the redundant features are removed after five iterations on cross validation. SWS on the other hand selects the optimum features by adding new features sequentially to the iteratively constructed regression model and evaluates the improvement on each new addition. This process happens iteratively until no further improvement can be provided by any newly added features.

2.3 Machine learning techniques

In this study, we use multivariate adaptive regression (MARS), support vector machine (SVM), random forest (RF), and extreme gradient boosting (XGBoost) for modeling. MARS is a nonlinear regression model improved from linear regression. The nonlinearity between input and targeted features can be captured by discrete splines (basis function) connected with nodes as shown on Fig. 3(a). MARS is good for its flexibility and interpretability but is sensitive to outliers. SVM is a powerful ML technique that can be used for both classification and regression. In classification, SVM finds the optimum hyperplane that maximizes the margins between different classes. In regression, SVM seeks the hyperplane that can capture the maximum variance between the input and targeted features as shown on Fig. 3(b). We use kernel function Radial Basis Function (RBF) to raise the dimension of the modeling space so that nonlinearities can be easily identified on modeling. Improved from basic decision tree, RF constructs multiple tree models together to form a tree forest, with each tree branching randomly to avoid any bias throughout the modeling. The prediction result of RF is given by averaging all the predictions of each tree as shown on Fig. 3(c). Another improved tree-based model, XGBoost builds trees sequentially to let new trees learn on the residual error of previous trees, so to improve the accuracy of prediction. The prediction result of XGBoost is obtained by adding the predictions of all trees as shown on Fig. 3(d). Both RF and XGBoost models belong to the ensemble learning method that combines the power of multiple simple ML models (decision tree models) to achieve better predictions.



Copyright © 2024 by ASME

Downloaded from http://asmedigitalcollection.asme.org/HT/proceedings-pdf/HT2024/87905/V001T03A003/3754451/v001t03a003-ht2024-131027.pdf by Tsinghua University user on 11 March 2026

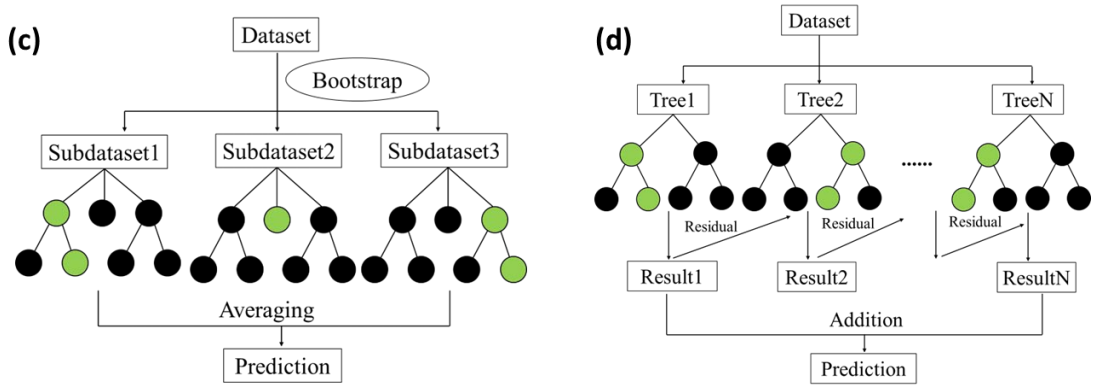


FIGURE 3: SCHEMATIC OF ML MODELS.

2.4 Zuber's model

To develop a model based on hydrodynamic instability, Zuber [2] first assumes that Taylor's instability (λ_t) happens between the interface of lifting bubbles and overhead surrounding liquid, as vapor bubbles patch together at the final stage of nucleate boiling on the transition to CHF. As bubbles continue leaving from surface, vapor jet forms at locations with the most dangerous Taylor's instability wavelength ($\lambda_{t,d}$) on the entire surface. As vapor jets flow to the top, the liquid surrounding these jet moves towards the bottom to quench the boiling surface and prevent it from reaching CHF. As further increase of the velocity of vapor jet, Helmholtz instability (λ_h) happens and the interface between the vapor jet and liquid becomes highly unstable. At CHF, the vapor jets merge to block the path for liquid from rewetting the surface, triggering the surge of wall superheat. Combining all these information, Zuber [2]'s correlation can be formulated as:

$$q''_{CHF} = K \cdot \rho_g h_{fg} \left(\frac{\sigma g \left(\rho_f - \rho_g \right)}{\rho_g^2} \right)^{1/4} \tag{3}$$

For decades, Zuber [2]'s correlation has been prevailing for pool boiling CHF of flat plate and multiplier K has been improved to incorporate important parameters relating to CHF. However, the improvement is limited by poor regression capability of the explicit function form and therefore ML models are required to unravel true relations on CHF. Fig. 4 shows the picture of Zuber [2]'s hydrodynamic instability model.

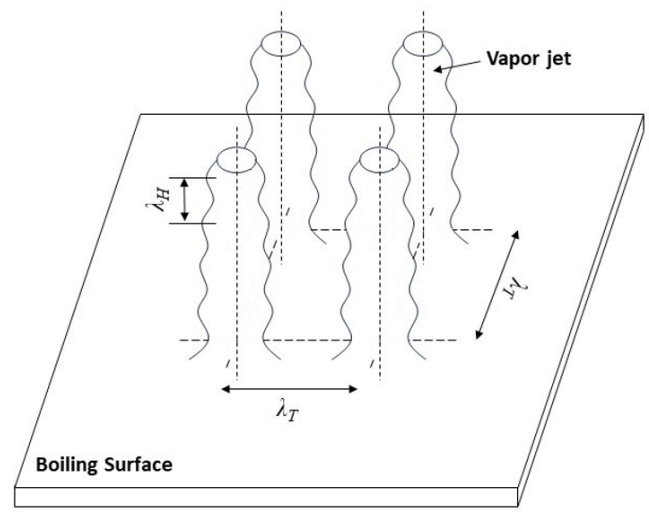


FIGURE 4: SCHEMATIC OF ZUBER'S MODEL

2.5 DDML vs. PAML models

In this study, we use both DDML and PAML frameworks to develop the ML models. For each framework, ML models are input by the features selected from feature selection methods (RFE and SWS), and the full feature set as well. A constructed model corresponds to single input and output. Therefore, a total of 12 models are constructed on each framework [(1 full feature set + 2 feature selection methods) × 4 ML models], and total 24 models are constructed throughout this work [12 models on each framework × 2 frameworks (DDML, PAML)]. For DDML, CHF predictions are given by the output of each ML model so that no additional information is involved during modeling. For PAML, instead of CHF as output, ML models train and test on the multiplier K of Zuber [2]'s hydrodynamic instability model with same group of input. The CHF is then recovered by multiplying the hydrodynamic instability term with predicted K as shown on Equation (3). Fig. 5 illustrates the DDML vs. PAML process in detail.

Downloaded from http://asmedigitalcollection.asme.org/HT/proceedings-pdf/HT2024/87905/V001T03A003/3754451/v001t03a003-ht2024-131027.pdf by Tsinghua University user on 11 March 2026

Copyright © 2024 by ASME

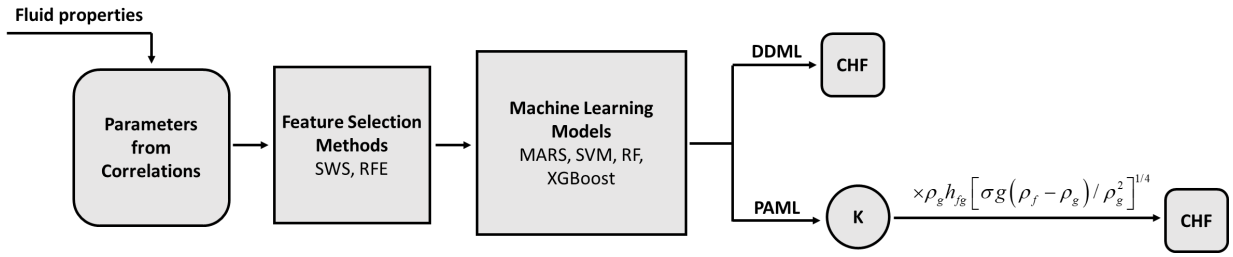


FIGURE 5: FLOW OF DDML VS. PAML.

From PAML framework, we expect additional information of physics could be incorporated from Zuber [2]'s model during modeling so that better predictions can be achieved by same group of input and models.

3. RESULTS AND DISCUSSION

3.1 Evaluation of traditional correlations

Before developing any ML models, we should understand the performance of traditional semi-empirical correlations on LHe. Two models are selected to test on the testing dataset to offer chance of comparison with the ML models developed in this study as shown on Table 2. The first model is introduced by Hata et al. [35], which incorporates the effects of orientation (θ) and subcooling (ΔT_{sub}). Hata et al. [35]'s model is selected as it was developed specifically for LHe with wide range of operational conditions ($0 < \Delta T_{sub} < 1.8$ K, $0.10 < P < 0.13$ MPa, and $0 < \theta < 180°$). The second model is by Kandlikar [37], who developed the model exclusively based on theoretical analysis. Kandlikar [37]'s model is chosen since we need to know the performance of correlation that is not empirically derived from LHe data, and since it has incorporated multiple parametric effects of CHF (orientation, contact angle). It should be noted that both models are improved hydrodynamic instability model of Zuber [2] with multipliers containing important parametric effects to CHF.

TABLE 2: TRADITIONAL CORRELATIONS FOR TESTING

Authors	Correlations
Hata et al. [35]	$q''_{CHF,sub} = \left\{\frac{1+0.065\left(\rho_f/\rho_g\right)^{0.8}\left(C_{pf}\Delta T_{sub}/h_{fg}\right)}{1+0.01\exp\left[2-0.33\left(C_{pf}\Delta T_{sub}/h_{fg}\right)\theta\right]}\right\}\times$ $0.17\rho_g h_{fg}[\sigma g(\rho_f-\rho_f)/\rho_g^2]^{1/4}\left(\frac{\rho_f}{\rho_f+\rho_g}\right)^{1.5}$
Kandlikar [37]	$q''_{CHF} = h_{fg}\rho_g^{1/2}\left(\frac{1+\cos\beta}{16}\right)\left[\frac{2}{\pi}+\frac{\pi}{4}(1+\cos\beta)\cos\theta\right]^{1/2}$ $\times[\sigma g(\rho_f-\rho_f)]^{1/4}$ $\beta = 20°$ *recommended for cryogenic fluids*

Fig. 6 shows plots of predicted versus actual CHF by Hata et al. [35] (Fig. 6(a)) and Kandlikar [34] (Fig. 6(b)) respectively. Hata et al. [35]'s model gives a reasonable prediction on CHF with 26.6% MAPE, compared to Kandlikar [37] that perform much worse with 52.8% MAPE. The results show that the traditional correlation model, even the one correlating on LHe, is not accurate enough to predict pool boiling CHF of LHe. Only reasonable prediction is expected on models correlating with LHe, while much worse predictions tend to be achieved on models without LHe data.

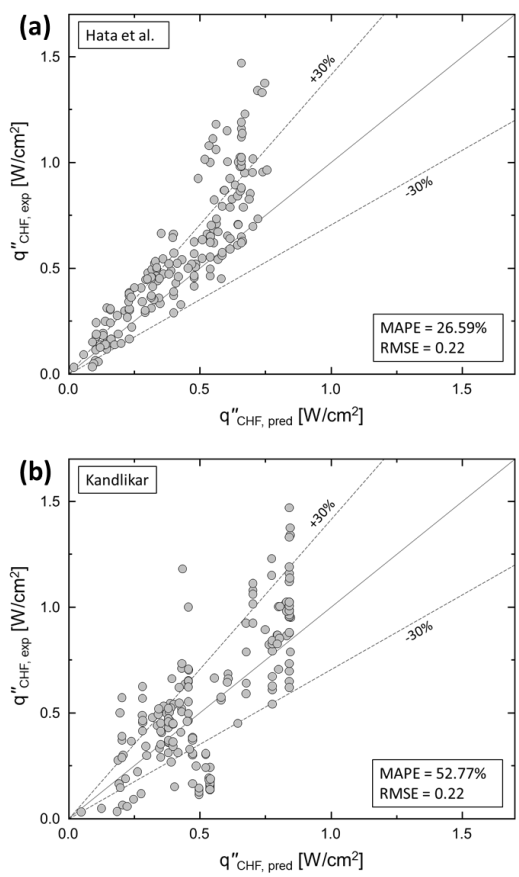


FIGURE 6: PERFORMANCE OF TRADITIONAL CORRELATIONS

Copyright © 2024 by ASME

Downloaded from http://asmedigitalcollection.asme.org/HT/proceedings-pdf/HT2024/87905/V001T03A003/3754451/v001t03a003-ht2024-131027.pdf by Tsinghua University user on 11 March 2026

3.2 Evaluation of DDML models

While Table 3 lists input features selected by each feature selection method, Table 4 lists the assessment of each model under these selected input features in both DDML and PAML framework. We first understand the assessment of ML models on DDML framework. From Table 4, the best MAPEs for each model across feature selection methods are 73.3%, 19.1%, 15.6%, and 15.7% for MARS, SVM, RF, and XGBoost respectively. Furthermore, when averaging over MAPEs of each model over feature selection methods, we have 110.8%, 20.9%, 15.7%, 15.9% averaged MAPEs for MARS, SVM, RF, and XGBoost. The results show that MARS gives much worse prediction, while SVM, RF, and XGBoost achieve better prediction compared to the tested traditional correlations [31,34]. This means that the discrete splines for MARS cannot offer an accurate capture of the CHF in terms of input features while the kernel function (RBF) for SVM, the tree models for RF and XGBoost do provide the ability for the models to accurately capture CHF of LHe. Furthermore, compared to SVM, RF and XGBoost result in more accurate predictions across each feature selection method. The reason is because both RF and XGBoost models are the ensemble learning method that combines the predictive power of multiple simpler models (decision trees) to achieve more accurate predictions. Another note is that the prediction performances across feature selection methods of each model are comparable except MARS in which a 111% difference is observed between SWS and full feature set. Fig. 7(a)–(d) shows the predicted versus actual CHF for all the best performed models across each feature selection method under DDML framework.

TABLE 3: SELECTED INPUT FEATURES

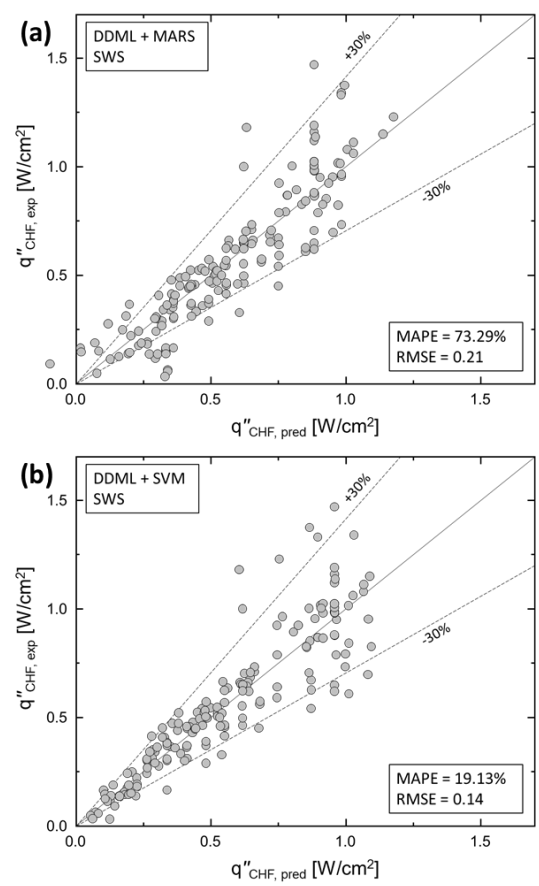
Frameworks	Feature Selection Methods	Selected Features
DDML	RFE	ρ_g, ρ_f, h_{fg}, σ, P, T_{sat}, k_f, ΔT_{sub}, θ
	SWS	ρ_g, h_{fg}, ΔT_{sub}, θ
PAML	RFE	ρ_f, h_{fg}, σ, P, ΔT_{sub}, θ
	SWS	ρ_g, ρ_f, h_{fg}, P, k_f, ΔT_{sub}, θ

TABLE 4: ML MODEL ASSESSMENT

Models	Frameworks	Feature Selection Methods	MAPE %	RMSE W/cm^2
MARS	DDML	RFE	74.51	0.218
		SWS	73.29	0.216
		FULL	184.59	0.647
	PAML	RFE	21.44	0.135
		SWS	22.42	0.136
		FULL	21.88	0.135
SVM	DDML	RFE	21.76	0.138
		SWS	19.13	0.141
		FULL	21.78	0.137
	PAML	RFE	17.21	0.137
		SWS	16.81	0.138
		FULL	16.82	0.135
RF	DDML	RFE	15.56	0.120
		SWS	15.87	0.121
		FULL	15.73	0.120
	PAML	RFE	15.62	0.121
		SWS	15.62	0.121
		FULL	15.51	0.121
XGBoost	DDML	RFE	15.72	0.112
		SWS	15.97	0.110
		FULL	16.04	0.111
	PAML	RFE	14.40	0.107
		SWS	14.29	0.109
		FULL	**14.03**	**0.106**

3.3 Evaluation of PAML models

As seen on Table 4, the performances of all ML models are boosted transition from DDML to PAML framework, and the averaged improvements over feature selection methods of each model are 88.9% (110.8% to 21.9%) for MARS, 4% (20.9 to 16.9%) for SVM, 0.14% (15.7% to 15.6%) for RF, and 1.7% (15.9% to 14.2%) for XGBoost, respectively. In addition, the best MAPEs across each feature selection method for each model are 21.4%, 16.8%, 15.5%, and 14% in PAML framework, compared to 73.3%, 19.1%, 15.6%, and 15.7% in DDML framework for MARS, SVM, RF, and XGBoost, respectively.



Downloaded from http://asmedigitalcollection.asme.org/HT/proceedings-pdf/HT2024/87905/V001T03A003/3754451/v001t03a003-ht2024-131027.pdf by Tsinghua University user on 11 March 2026

Copyright © 2024 by ASME

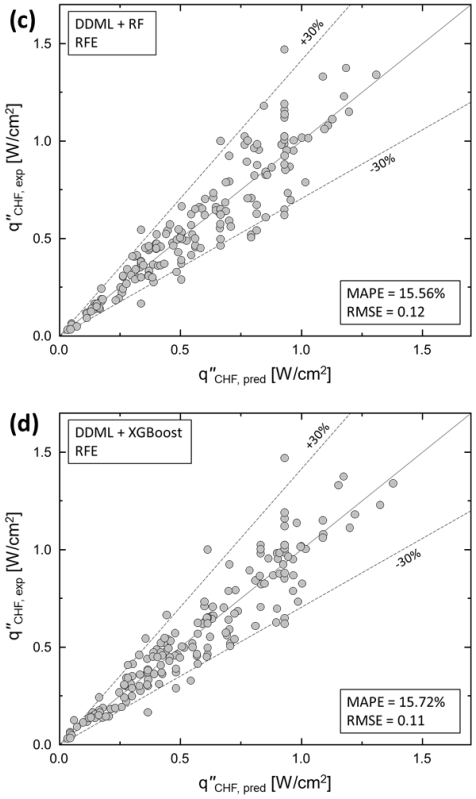
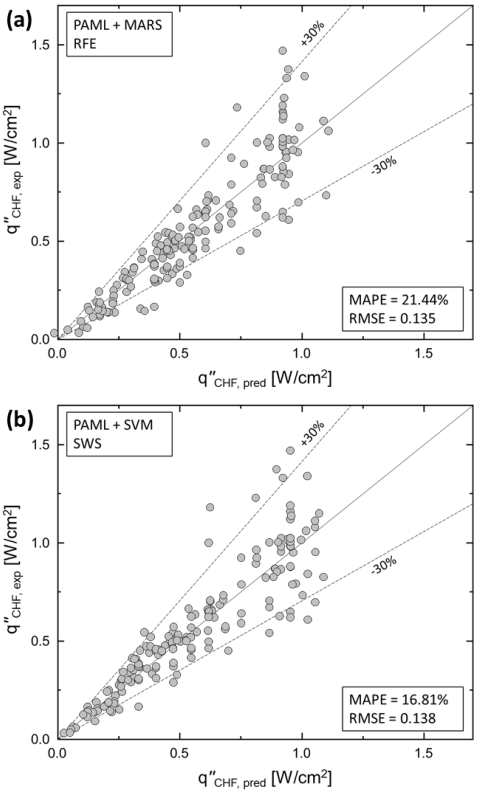


FIGURE 7: PERFORMANCE OF ML MODELS ON DDML FRAMEWORK



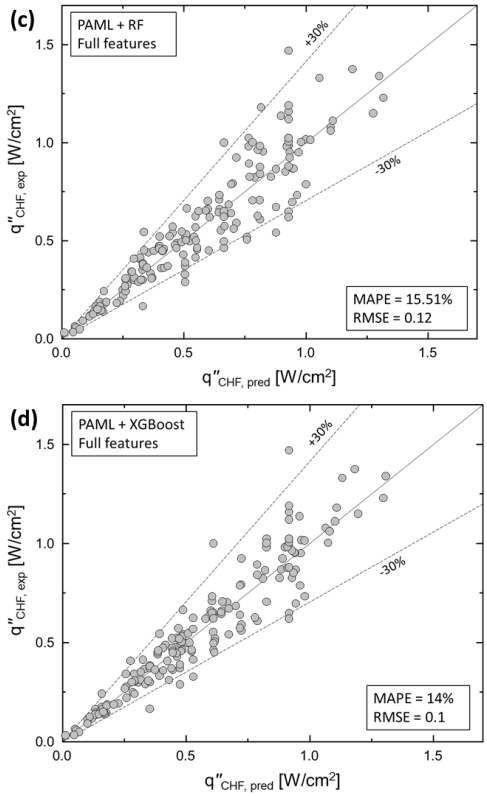


FIGURE 8: PERFORMANCE OF ML MODELS ON PAML FRAMEWORK

Among them, XGBoost on PAML framework performs the best with 14% MAPE. In result, we conclude that the transition from DDML to PAML framework gives the most dramatic improvement on MARS, reasonable improvement on SVM, and little improvement on XGBoost and RF models. We should note that all the improvements result from the assistance of Zuber [2]'s model that offers additional information of physics in modeling, enabling an accurate capture on CHF of LHe under wide range of operational conditions. We should also note that less improvement on RF and XGBoost may result from the fact that the predictive power of multiple models of ensemble learning method are enough to unravel the intricate patterns of CHF over input features in this study, limiting the assistance offered by Zuber [2]'s model. Fig. 8(abcd) shows the predicted versus actual CHF for all the best performed ML models across each feature selection method under PAML framework.

4. Conclusion

A physics-assisted machine learning framework is proposed to enable accurate prediction of pool boiling CHF of LHe. This is achieved first by amassing a large dataset of CHF on LHe from worldwide published literatures. Multivariate adaptive regression, support vector machine, random forest, and extreme gradient boosting are used for modeling while recursive feature elimination and stepwise selection are adapted to remove irrelevant features before modeling. With selected machine learning models tested on both traditional and physics-assisted

Downloaded from http://asmedigitalcollection.asme.org/HT/proceedings-pdf/HT2024/87905/V001T03A003/3754451/v001t03a003-ht2024-131027.pdf by Tsinghua University user on 11 March 2026

Copyright © 2024 by ASME

frameworks, conclusions are: (i) best performed model is achieved by extreme gradient boosting on physics-assisted framework with 14% mean absolute error, (ii) physics-assisted framework can greatly enhance the predictive power of multivariate adaptive regression and support vector machines, while having less effect on random forest and extreme gradient boosting (models with ensemble learning method), (iii) random forest and extreme gradient boosting models can greatly outperform the traditional semi-empirical correlations in both frameworks.

ACKNOWLEDGMENTS

The authors are grateful for the financial support by National Aeronautics and Astronautics Administration (NASA) under Grant Number: 80NSSC21C0089 and Grant Number: 80NSSC23CA010 of STTR Phase I and II programs.

REFERENCES

[1] Liang, G., and Mudawar, I., 2018, "Pool Boiling Critical Heat Flux (CHF) – Part 1: Review of Mechanisms, Models, and Correlations," Int J Heat Mass Transf, **117**, pp. 1352–1367.

[2] Zuber, 1958, "Further Remarks on the Stability of Boiling Heat Transfer," Physics and Mathematics.

[3] Wang, L., Li, Y., Zhang, F., Xie, F., and Ma, Y., 2016, "Correlations for Calculating Heat Transfer of Hydrogen Pool Boiling," Int J Hydrogen Energy, **41**(38), pp. 17118–17131.

[4] Bewilogua, L., Knoner, R., and Vinzelberg, H., 1975, *Heat Transfer in Cryogenic Liquids under Pressure*.

[5] Chang, J. Y., and You, S. M., 1996, "Heater Orientation Effects on Pool Boiling of Micro-Porous-Enhanced Surfaces in Saturated FC-72."

[6] Ivey, H. J., 1962, "On the Relevance of the VaporLiquid Exchange Mechanism for Subcooled Boiling Heat Transfer at High Prossure," UKAEA.

[7] Qiu, Y., Garg, D., Zhou, L., Kharangate, C. R., Kim, S.-M., and Mudawar, I., 2020, "An Artificial Neural Network Model to Predict Mini/Micro-Channels Saturated Flow Boiling Heat Transfer Coefficient Based on Universal Consolidated Data," Int J Heat Mass Transf, **149**, p. 119211.

[8] Kim, S.-M., and Mudawar, I., 2013, "Universal Approach to Predicting Saturated Flow Boiling Heat Transfer in Mini/Micro-Channels–Part II. Two-Phase Heat Transfer Coefficient," Int J Heat Mass Transf, **64**, pp. 1239–1256.

[9] Zhou, L., Garg, D., Qiu, Y., Kim, S.-M., Mudawar, I., and Kharangate, C. R., 2020, "Machine Learning Algorithms to Predict Flow Condensation Heat Transfer Coefficient in Mini/Micro-Channel Utilizing Universal Data," Int J Heat Mass Transf, **162**, p. 120351.

[10] Kim, S.-M., and Mudawar, I., 2013, "Universal Approach to Predicting Heat Transfer Coefficient for Condensing Mini/Micro-Channel Flow," Int J Heat Mass Transf, **56**(1–2), pp. 238–250.

[11] Bahman, A. M., and Ebrahim, S. A., 2020, "Prediction of the Minimum Film Boiling Temperature Using Artificial Neural Network," Int J Heat Mass Transf, **155**, p. 119834.

[12] Ravichandran, M., and Bucci, M., 2019, "Online, Quasi-Real-Time Analysis of High-Resolution, Infrared, Boiling Heat Transfer Investigations Using Artificial Neural Networks," Appl Therm Eng, **163**, p. 114357.

[13] Kim, K. M., Hurley, P., and Duarte, J. P., 2022, "Physics-Informed Machine Learning-Aided Framework for Prediction of Minimum Film Boiling Temperature," Int J Heat Mass Transf, **191**, p. 122839.

[14] Li, J., O'Neill, L. E., Izenson, M. G., and Kharangate, C. R., 2023, "Data Consolidation, Correlations Assessment, and New Correlation Development for Pool Boiling Critical Heat Flux Specific to Cryogenic Fluids," Int J Heat Mass Transf, **213**, p. 124315.

[15] Lemmon, E. W., Huber, M. L., and McLinden, M. O., 2010, "NIST Standard Reference Database 23," Reference fluid thermodynamic and transport properties (REFPROP), version, **9**.

[16] Lyon, D. N., 1965, "Boiling Heat Transfer and Peak Nucleate Boiling Fluxes in Saturated Liquidhelium between the Ramda and Critical Temperatures," Adv. Cryog. Eng., **10**, p. 371.

[17] R.D. Cummings, 1966, "Boiling Heat Transfer to Liquid Helium," Pure and Applied Cryogenics, pp. 85–95.

[18] Jergel, M., and Stevenson, R., 1971, "Static Heat Transfer to Liquid Helium in Open Pools and Narrow Channels," Int J Heat Mass Transf, **14**(12), pp. 2099–2107.

[19] Jergel, M., and Stevenson, R., 1972, *HEAT TRANSFER TO BOILING HELIUM FROM ALUMINIUM SURFACES.*, Electrotechnical Inst., Bratislava.

[20] Jergel, M., and Stevenson, R., 1974, "Contribution to the Static Heat Transfer to Boiling Liquid Helium," Cryogenics (Guildf), **14**(8), pp. 431–433.

[21] Tsukamoto, O., and Kobayashi, S., 1975, "Transient Heat Transfer Characteristics of Liquid Helium," J Appl Phys, **46**(3), pp. 1359–1364.

[22] Vishnev, Ip., YG, V., and GG, S., 1976, "STUDY OF HEAT TRANSFER IN BOILING OF HELIUM ON SURFACES WITH VARIOUS ORIENTATIONS."

[23] Deev, V. I., Keilin, V. E., Kovalev, I. A., Kondratenko, A. K., and Petrovichev, V. I., 1977, "Nucleate and Film Pool Boiling Heat Transfer to Saturated Liquid Helium," Cryogenics (Guildf), **17**(10), pp. 557–562.

[24] Grigoriev, V. A., Klimenko, V. V, Pavlov, Y. M., and Ametistov, E. V, 1978, "The Influence of Some Heating Surface Properties on the Critical Heat Flux in Cryogenic Liquids Boiling," *International Heat Transfer Conference Digital Library*, Begel House Inc.

[25] Ibrahim, E. A., Boom, R. W., and McIntosh, G. E., 1978, "Heat Transfer to Subcooled Liquid Helium," *Advances in Cryogenic Engineering*, Springer, pp. 333–339.

Downloaded from http://asmedigitalcollection.asme.org/HT/proceedings-pdf/HT2024/87905/V001T03A003/3754451/v001t03a003-ht2024-131027.pdf by Tsinghua University user on 11 March 2026

Copyright © 2024 by ASME

[26] Tsuruga, H., and Endoh, K., 1980, "Pool Boiling Heat Transfer to Saturated Liquid Helium," TEION KOGAKU (Journal of Cryogenics and Superconductivity Society of Japan), **15**(5), pp. 283–287.

[27] Verkin, B. I., Kirichenko, Y. A., Kozlov, S. M., and Rusanov, K. V, 1980, "Heat Transfer during Pool Boiling of Subcooled Helium," *Proc 8th Intern Cryog Eng Conference*, pp. 256–260.

[28] Ogata, H., and Nakayama, W., 1982, "Heat Transfer to Boiling Helium from Machined and Chemically Treated Copper Surfaces."

[29] Caspi, S., 1984, *Heat Transfer to Subcooled He I*, Springer.

[30] NISHIO, S., and CHANDRATILLEKE, G. R., 1989, "Steady-State Pool Boiling Heat Transfer to Saturated Liquid Helium at Atmospheric Pressure," JSME international journal. Ser. 2, Fluids engineering, heat transfer, power, combustion, thermophysical properties, **32**(4), pp. 639–645.

[31] Ogata, H., and Mori, H., 1993, "Steady State Heat Transfer in Transition Boiling of Helium on Copper Surfaces," Cryogenics (Guildf), **33**(6), pp. 640–642.

[32] CHANDRATILLEKE, G. R., and Nishio, S., 1988, "Pool Boiling Heat Transfer to Saturated Liquid Helium on Coated Surfaces," TEION KOGAKU (Journal of Cryogenics and Superconductivity Society of Japan), **23**(3), pp. 128–133.

[33] Iwamoto, A., Mito, T., Takahata, K., Yanagi, N., and Yamamoto, J., 1996, "Heat Transfer from an Oxidized Large Copper Surface to Liquid Helium: Dependence on Surface Orientation and Treatment," *Advances in Cryogenic Engineering*, Springer, pp. 217–224.

[34] Iwamoto, A., Maekawa, R., Mito, T., and Satoh, S., 1998, "Heat Transfer Characteristics of Liquid Helium with Different Surface Width and Length," *International Cryogenic Engineering Conference*, pp. 801–804.

[35] Hata, K., 2003, "Critical Heat Flux on a Flat Plate in a Pool of Subcooled Liquid Helium," AIP Publishing, pp. 1460–1467.

[36] Iwamoto, A., Maekawa, R., and Mito, T., 2006, "Steady State Heat Transfer of an Oxidized Copper Surface in Subcooled Liquid Helium," Fusion Engineering and Design, **81**(23–24), pp. 2611–2615.

[37] Kandlikar, S. G., 2001, "A Theoretical Model to Predict Pool Boiling CHF Incorporating Effects of Contact Angle and Orientation," J. Heat transfer, **123**(6), pp. 1071–1079.

Downloaded from http://asmedigitalcollection.asme.org/HT/proceedings-pdf/HT2024/87905/V001T03A003/7375451/v001t03a003-ht2024-131027.pdf by Tsinghua University user on 11 March 2026

Copyright © 2024 by ASME